Track 15: Vision-Language Alignment with CLIP for Video

Deep Learning: Advanced Models and Methods (DL26)

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https://github.com/weiss25r/Vision-Language-Alignment-with-CLIP-for-Video

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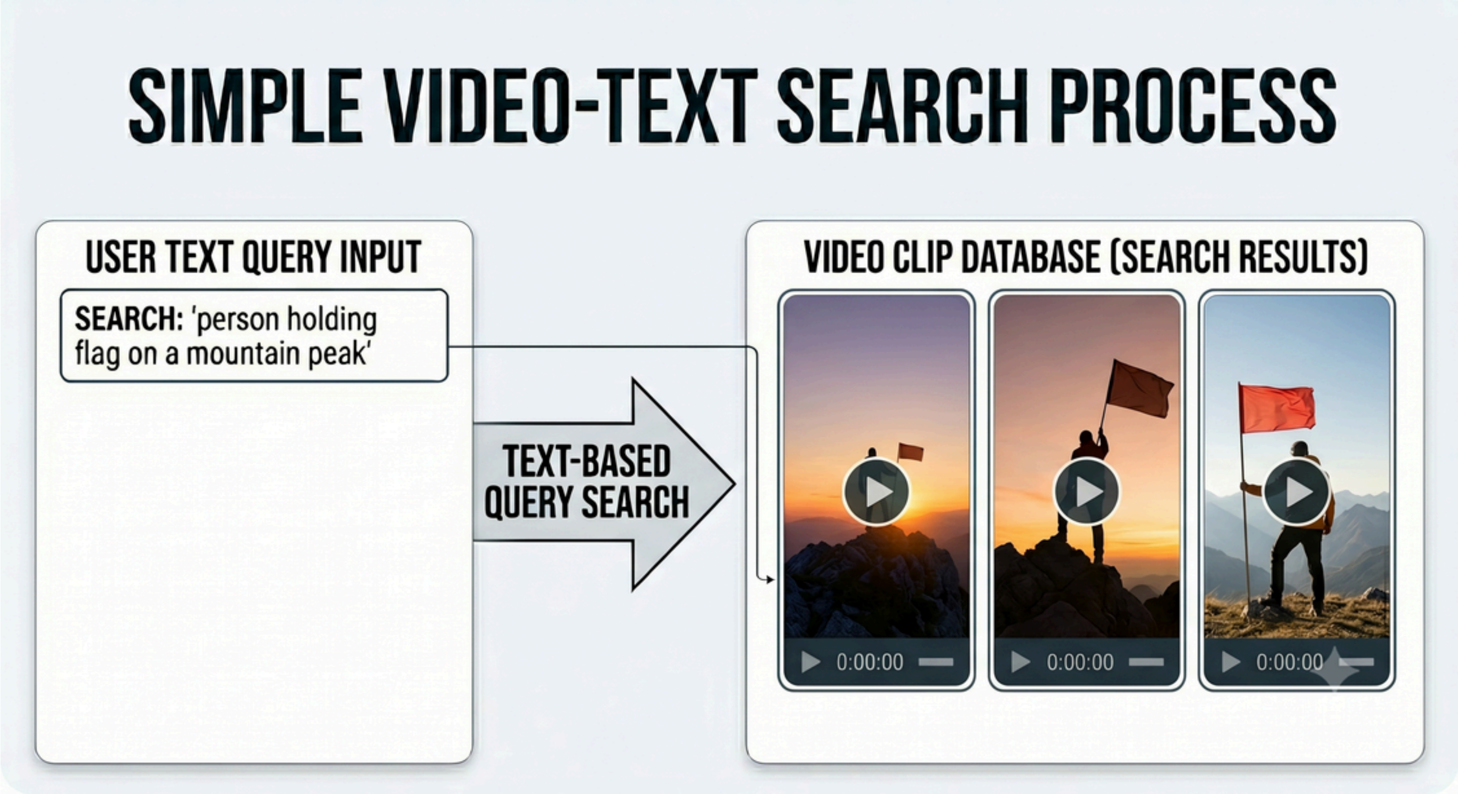
The Problem: Video Search

How do you find a video clip you've never labeled?

- If we have hours of video with no manual annotations, no searchable metadata
- Traditional retrieval: lable-based, brittle, does not scale
- Our goal: given a sentence like "person holding flag on a mountain peak", retrieve the right clip — zero labels, zero fine-tuning on the target domain



The EPIC KITCHENS dataset

Exploring Sampled Epic Kitchens 100

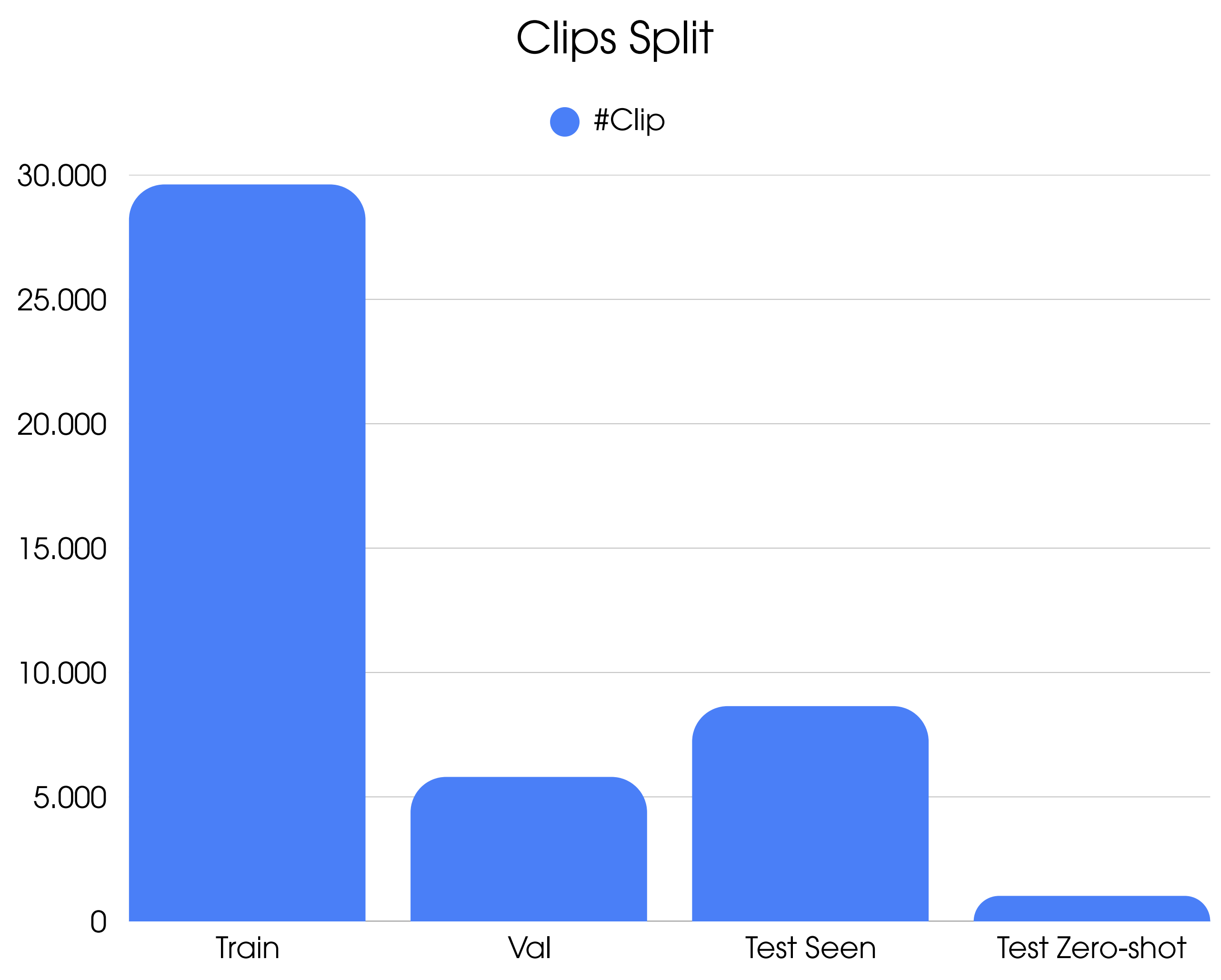
- 438 egocentric videos, 45K+ clip action segments, 90+ verb classes
- First-person kitchen activities: cutting, washing, pouring, stirring...
- Task: given a text query, rank all video clips by similarity

Experimental setup

- Hardware: NVIDIA RTX 4090
- Framework: PyTorch Lightning

Dataset Splits

- Train / Val — 85% / 15% of the annotated segments, used for training and hyperparameter tuning
- Test Seen — clips containing verbs and nouns observed during training, unseen narrations
- Test Zero-Shot — clips with verbs or nouns never seen during training — unseen narrations



Methodology

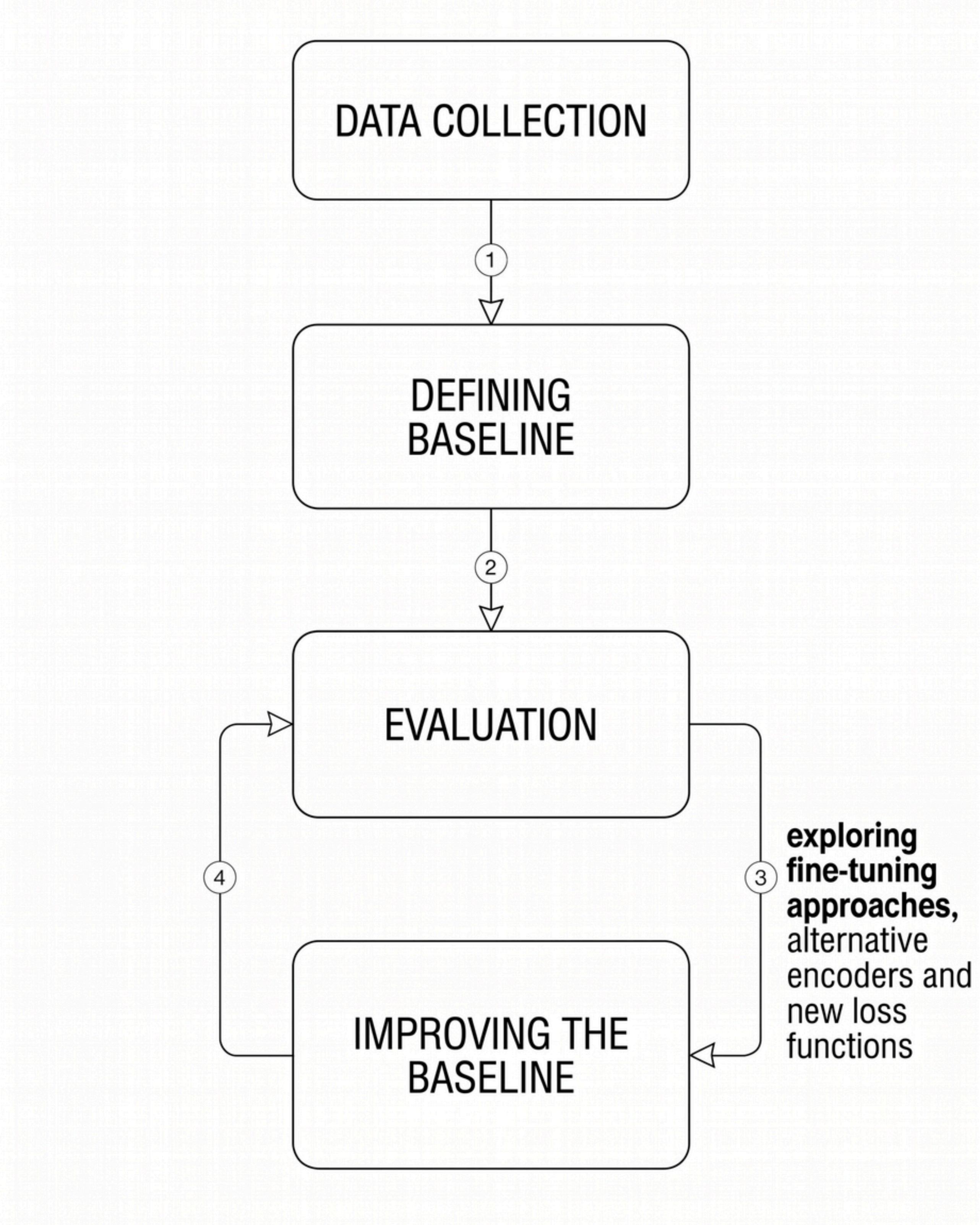
- <u>Defining Baseline</u>
  - Frozen video and text encoders MLP adapter with CLIP-style contrastive loss

- <u>Evaluation</u>
  - To perform model selection and evaluate our models, we compute Multi-Instance Recall

$$\mathrm{match}(i,j) = \mathbf{1}[\mathrm{verb}_i = \mathrm{verb}_j] \wedge \mathbf{1}[\mathrm{noun}_i = \mathrm{noun}_j]$$

$$\mathrm{MIR@K} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left[\exists\, j \in \mathrm{top\text{-}K}(i) : \mathrm{match}(i,j) = 1\right]$$

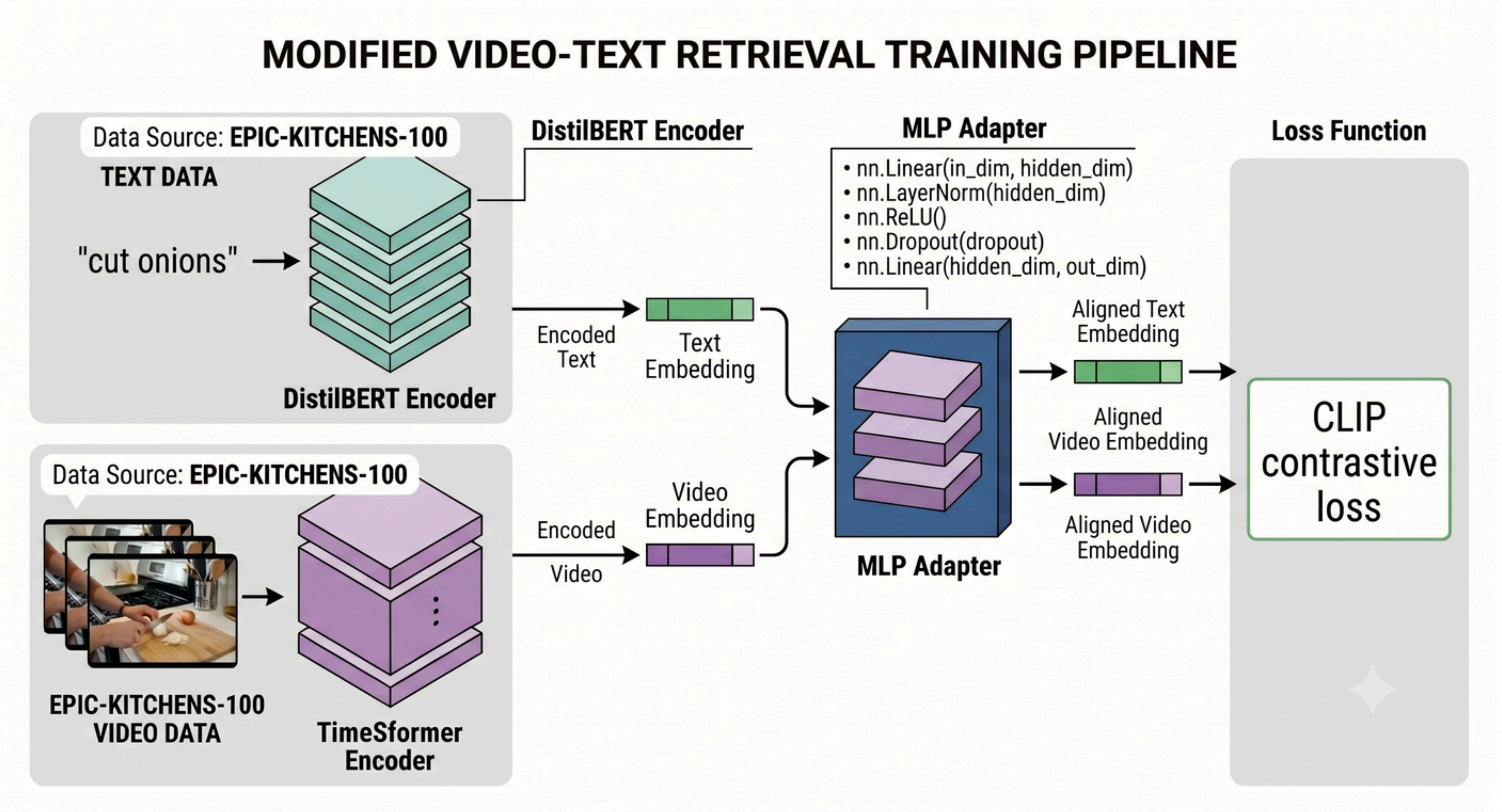
- <u>Improving the Baseline:</u>
  - Fine tuning, changing encoders and new loss functions



Baseline Model Overview

DistilBERT + TimeSformer MLP

- Text encoder: DistilBERT — lightweight BERT variant, frozen
- Video encoder: TimeSformer pre-extracted features — frozen
- Alignment: MLP adapter (1 hidden layer) trained with CLIP contrastive loss



Sanity Checks

- How representative are the video features ? To answer, before training, we perform two sanity checks:
  - zero shot evaluation - no adapter training
  - training a logistic classifier on verbs

Fine-tuning: DistilBERT + TimeSformer + linear adapter

Hypothesis: jointly fine-tuning text and video encoders improves alignment

- Setup:
  - Text: DistilBERT fine-tuned end-to-end
  - Video: TimeSformer last layer unfrozen
  - Linear adapter to reduce the computational load

- What we learned:
  - Unfreezing the last TSF layer degrades performance
  - Fine-tuning at this scale destroy feature representations
  - Pre-trained features carry more signal

Stronger Backbones: CLIP and EgoVLP

CLIP frozen + MLP:

- Text: CLIP text encoder — frozen
- Video: CLIP visual encoder ViT-B/32 — frozen
- What we learned:
  - CLIP underperforms despite stronger pre-training — egocentric video is out-of-distribution for a model trained on web images

EgoVLP + MLP

- Pre-trained on Ego4D, fine-tuned on EPIC-KITCHENS
- egocentric domain priors built in Two loss variants: standard CLIP loss vs EgoNCE loss
- What we learned:
  - Egocentric pre-training closes the domain gap — EgoNCE loss pushes alignment further by handling verb-object ambiguity native to first-person video.

Changing the Loss

The problem with standard CLIP loss:

- Treats each video-text pair as the only positive — all others are negatives
- In Epic Kitchens multiple clips depict the same action ("put down the knife")
- Semantically equivalent samples penalized as negatives

Our solution — action-aware positive sampling:

- Define positive set as all batch samples sharing same verb AND noun
- Loss distributes probability mass over the entire positive set instead of a single pair

$$\mathcal{L}_{v2t}^{\text{ego}} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\sum_{k\in\mathcal{P}_i} \exp(\mathbf{v}_i^T \mathbf{t}_k/\tau)}{\sum_{j=1}^{N} \exp(\mathbf{v}_i^T \mathbf{t}_j/\tau)}$$

where $\mathcal{P}_i = \{j \in \mathcal{B} \mid \mathrm{verb}(j) = \mathrm{verb}(i) \ \wedge \ \mathrm{noun}(j) = \mathrm{noun}(i)\}$

Final loss is computed as: $\frac{1}{2}\left(\mathcal{L}_{v2t}^{ego} + \mathcal{L}_{t2v}^{ego}\right)$

Egocentric Pre-training wins!

Input Features	MIR@1 (ZS)	MIR@5 (ZS)	MIR@10 (ZS)	accuracy (verbs)
TMSF + DBRT	0.1	1.4	3.6	32
CLIP	5.5	20.9	31.2	28
EgoVLP+	52.6	79.2	85.8	77

Sanity checks, results on validation set

Experiment	MIR@1	MIR@5	MIR@10
Baseline	24.6	57.7	66.1
Fine-tuning DBRT + TSF(last L) + Linear	25.4	54.6	66.1
CLIP frozen + MLP	21.0	44.3	55.0
EgoVLP + MLP (CLIP loss)	64.7	84.6	89.2
EgoVLP + MLP (EgoNCEloss)	68.9	84.1	88.2

Model Selection on validation set

Qualitative Analysis - Failure cases

Best Model Results

Set	MIR@1	MIR@5	MIR@10
Test Seen	52.4	70.3	76.3
Test zero-shot	49.6	73.2	81.2

Three failure categories:

- Semantically close — high similarity (≥0.70), model retrieves plausible alternatives — errors a human might consider acceptable
- Verb confusion — correct object, wrong fine-grained motion (take vs squeeze, wipe vs clean)
- Complete failures — low similarity (≤0.52), ambiguous or visually cluttered clips

Top-1 misclassifications on test seen

Query (GT)	Predicted (top1)	Cosine Sim
wipe cooker	clean kitchen	0.704
put down cloth	hang up cloth	0.714
throw can into bin	put rubbish in bin	0.618
hand cloth	squeeze cloth	0.619
take hand	pick up ball of dough	0.532
throw onion in	throw eggs box	0.515

Conclusions, Limitations & Future Work

Conclusions:

- Features quality dominates adapter complexity — domain-specific pre-training matters more than downstream architecture: +44% compared to the baseline

Limitations:

- Dataset sampling — we worked on a sampled subset, which may underrepresent rare verb-noun combinations
- full end-to-end fine-tuning was infeasible due to VRAM constraints on a single GPU

Future work:

- With multi-GPU setup, full fine-tuning could push R@1 significantly beyond 68.9
- Training on the complete Epic Kitchens without sampling would improve coverage of rare verb-noun pairs

Thank you for your attention!